جامعة أم القرى
UMM AL-QURA UNIVERSITY

Database Project Report

Title: Café Management System

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Team Responsibilities

Ghufran Talal Simbawah Phase:

Conceptual Design & Documentation. Tasks:Defining the Project Description and objectives. Identifying Entities, Attributes, and Relationships. Designing the ER Diagram. Writing the Conclusion.

Fatmah Obaid Alruways Phase:

Database Foundation & Logic.
Tasks:Converting the ER Diagram into a Relational Schema. Writing the Data Definition Language (DDL) for creating the database and tables. Applying all Data Integrity Constraints (PK, FK, Unique, Check). Handling the initial Data Entry (Insertion of 5 sample records per table).

Lujain Mohammed Alabdali Phase:

SQL Analysis & Reporting (The Presentation Part). Tasks:Writing and testing all specific SQL Queries (Basic Select, Aggregate, Joins, and Subqueries). Designing and creating the Order_Report View for automated reporting. Managing the "Appendix" by capturing and labeling all 13 required screenshots from MySQL Workbench. Final review of the report to ensure all query results match the database structure.

Part 1 Conceptual and Logical Design

Identify Entities Attributes and Relationships

1. Project Description

The Cafe Management System is designed to manage customers, menu items, and orders in a cafe. The system helps store customer information, menu details, and order records. It allows the cafe to track which customers made orders and which items were ordered.

2. Entities and Attributes

1. Customer
This entity stores information about customers.
Attributes:
• Customer_ID (Primary Key)
• Customer_Name
• Phone_Numbe
_____-

2. Item
This entity stores information about cafe menu items.
Attributes:
• Item_ID (Primary Key)
• Item_Name
• Price
_____

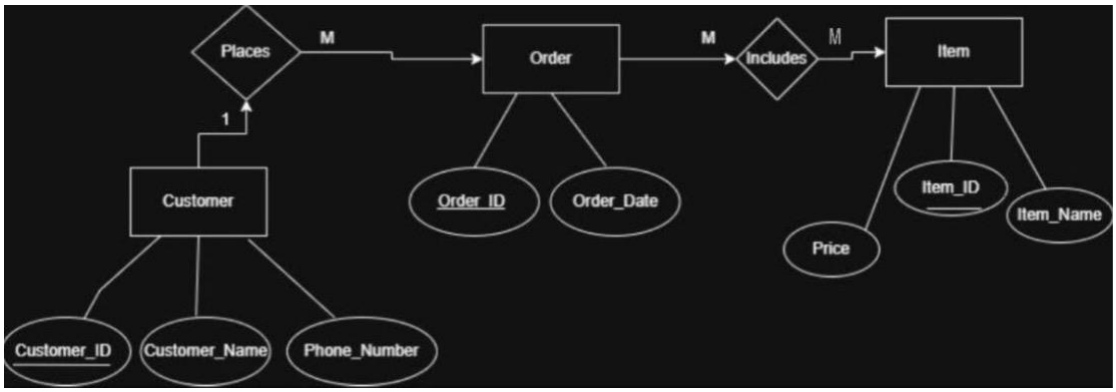
3. Order
This entity stores information about orders.
Attributes:
• Order_ID (Primary Key)
• Order_Date
• Customer_ID (Foreign Key)
• Item_ID (Foreign Key

3. Relationships

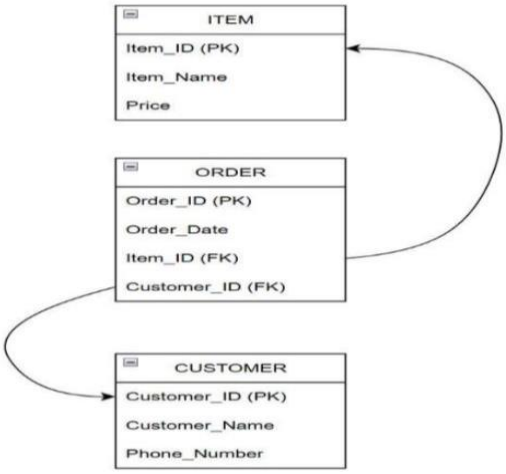
Relationship 1: Customer places Order
• One customer can place many orders.
• Each order belongs to one customer.
• Relationship type: One-to-Many (1:M)

Relationship 2: Item included in Order
• One item can appear in many orders.
• Each order contains one item.
• Relationship type: Many-to-Many (M:N)

ER Diagram



Relational Schema



Part 2 Database Implementation

Database Creation:
The database CafeManagementDB was created to provide a structured environment for managing the cafe's data.

Tables Creation & Data Types:
Three main tables were implemented:

1-CUSTOMER: Stores client data using INT for IDs and VARCHAR for names and contact details.
2-ITEM: Manages the menu items, utilizing the DECIMAL data type for the Price column to ensure financial accuracy.
3- ORDERS: Records transactions by linking customers to items using DATE to track when each order was placed.

Constraints:
To ensure data integrity, several constraints were applied:

-Primary Keys (PK): Defined for Customer_ID, Item_ID, and Order_ID to uniquely identify each record.
-Foreign Keys (FK): Implemented in the ORDERS table to maintain relationships between customers and items.
-Check & Unique: A CHECK (Price > 0) constraint ensures prices are positiv,while the UNIQUE constraint prevents duplicate phone numbers.

Sample Data:
Five sample records were inserted into each table (CUSTOMER, ITEM, and ORDERS) to populate the system for testing and demonstration.

Part 3 SQL Queries and Report

1- Basic SELECT Query 1:

-Purpose: Retrieves the names and phone numbers of customers whose phone numbers start with '0505'.

-Logic: Uses the LIKE operator for pattern matching.

2- Basic SELECT Query 2:

-Purpose: Lists all menu items with a price greater than 15.

-Logic: Uses the WHERE clause with a comparison operator.

3- Aggregate Function Query 1 (COUNT):

-Purpose: Calculates the total number of orders placed by each customer.

-Logic: Uses COUNT(*) along with the GROUP BY clause.

4- Aggregate Function Query 2 (AVG):

-Purpose: Finds the average price of all items on the menu.

-Logic: Applies the AVG() function to the Price column.

5- Join Query:

-Purpose: Generates a comprehensive report showing the Customer Name, Item Name, and Order Date.

-Logic: Performs a multiple JOIN operation between the ORDERS, CUSTOMER, and ITEM tables.

6- Subquery:

-Purpose: Identifies customers who ordered items priced above the overall average menu price.

-Logic: Utilizes nested SELECT statements to filter data based on calculated values.

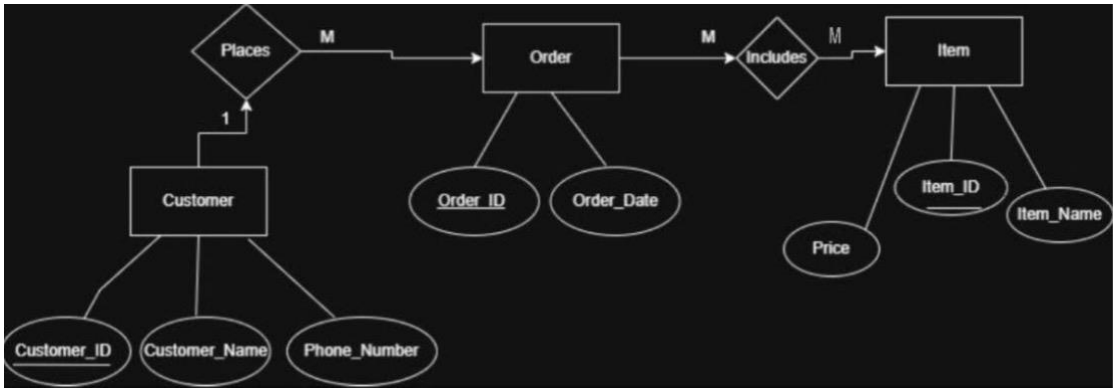
7- View:

-Purpose: Creates a virtual table named Order_Report to simplify the generation of future sales reports without re-writing complex join logic.

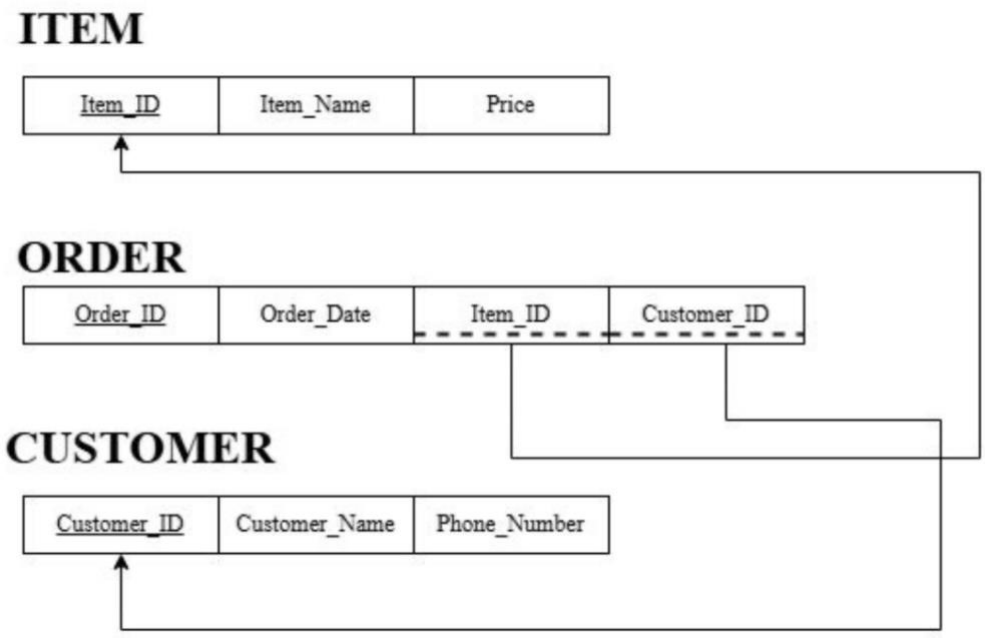
-Logic: Uses the CREATE VIEW statement to store the query structure.

Screenshots (Appendix)

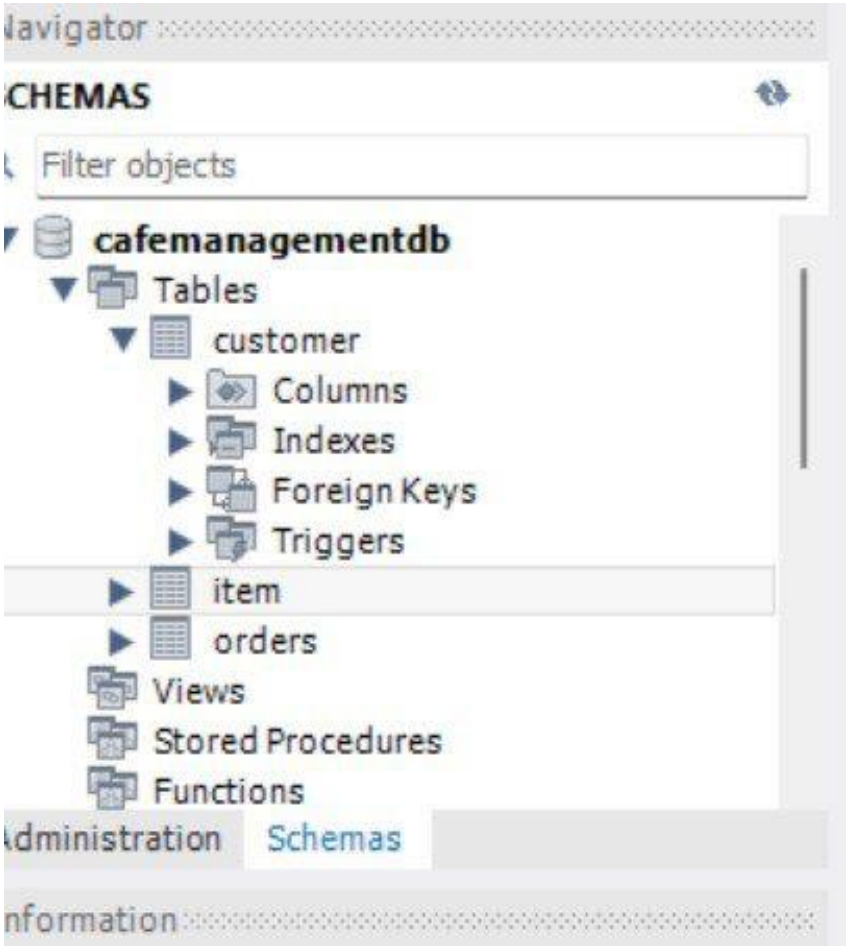
Screenshot 1: ER Diagram



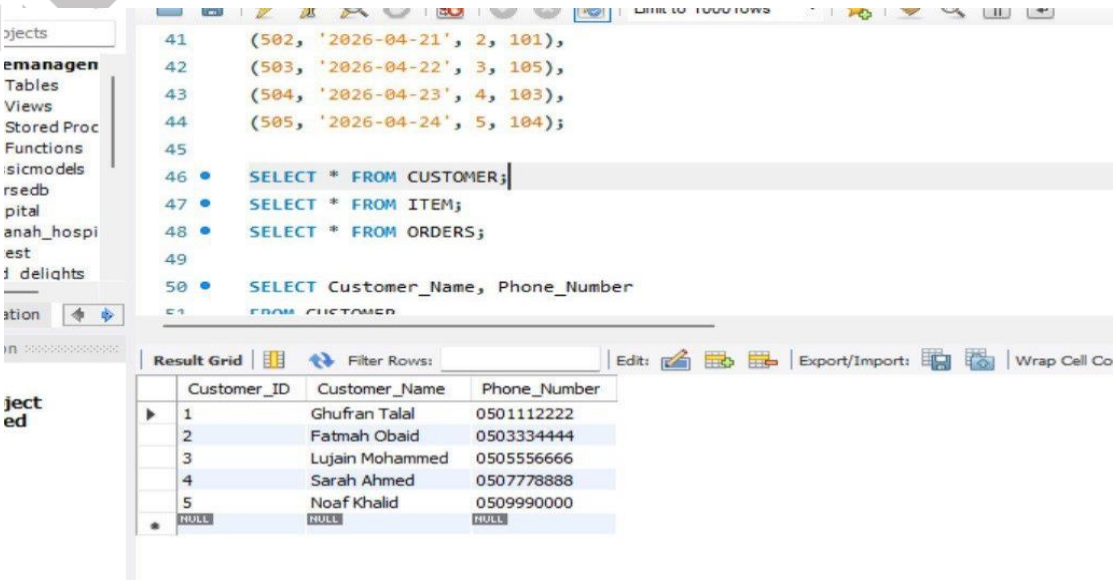
Screenshot 2: Relational Schema



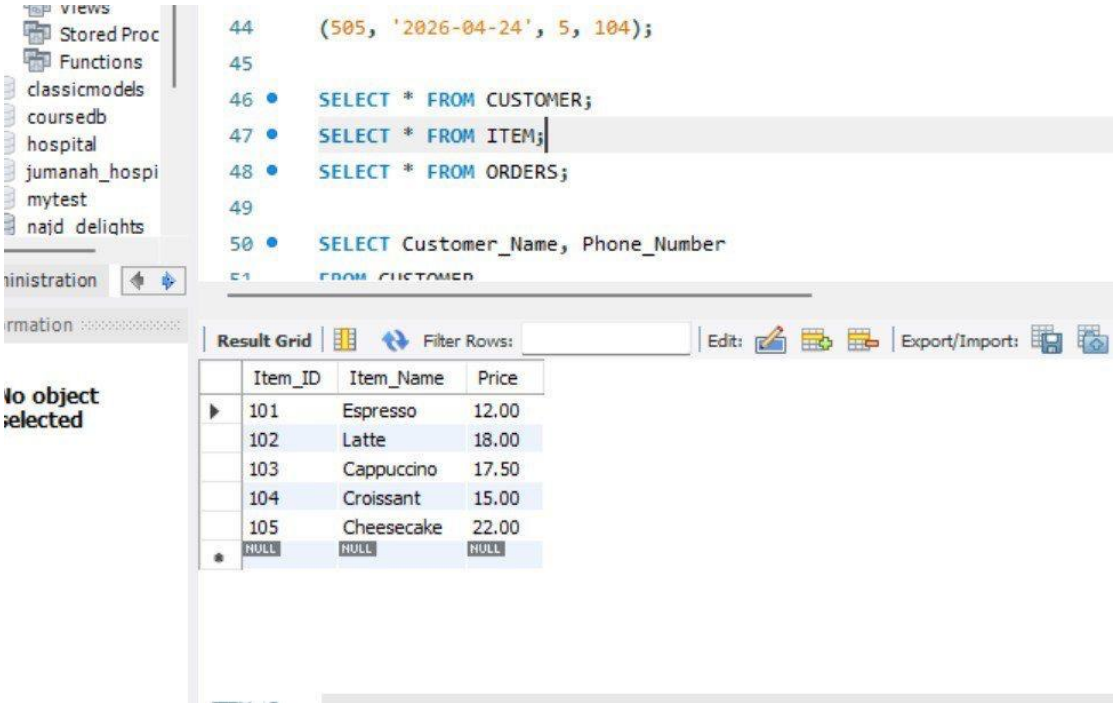
Screenshot 3,4 : Database Creation, Tables Structure



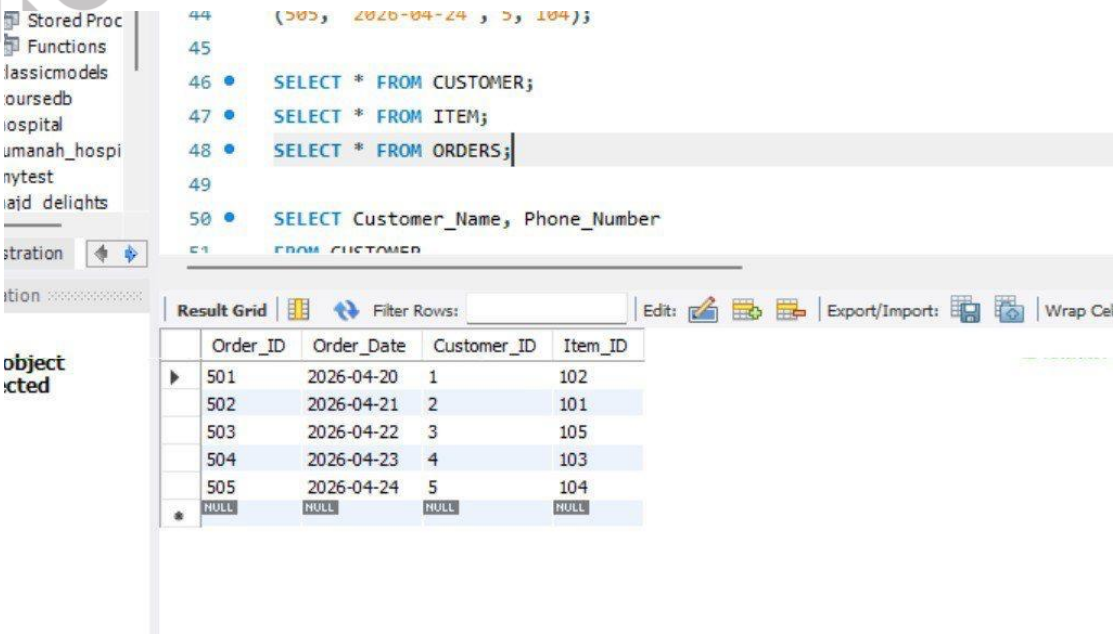
Screenshot 5: Sample Data (Table 1)



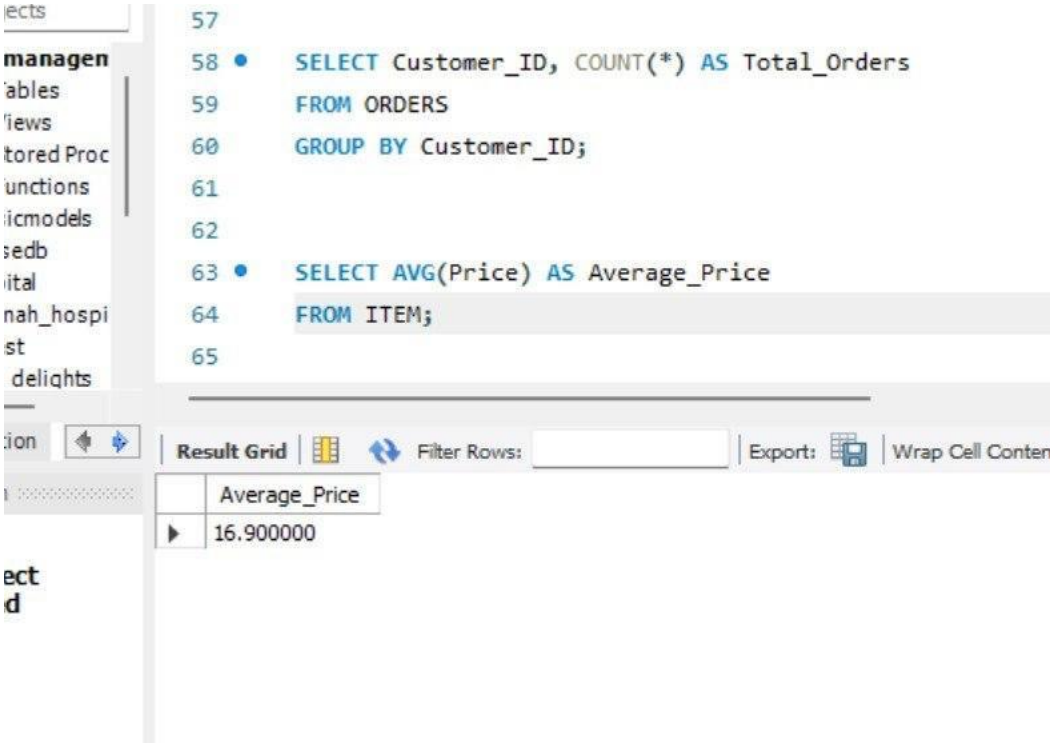
Screenshot 6: Sample Data (Table 2)



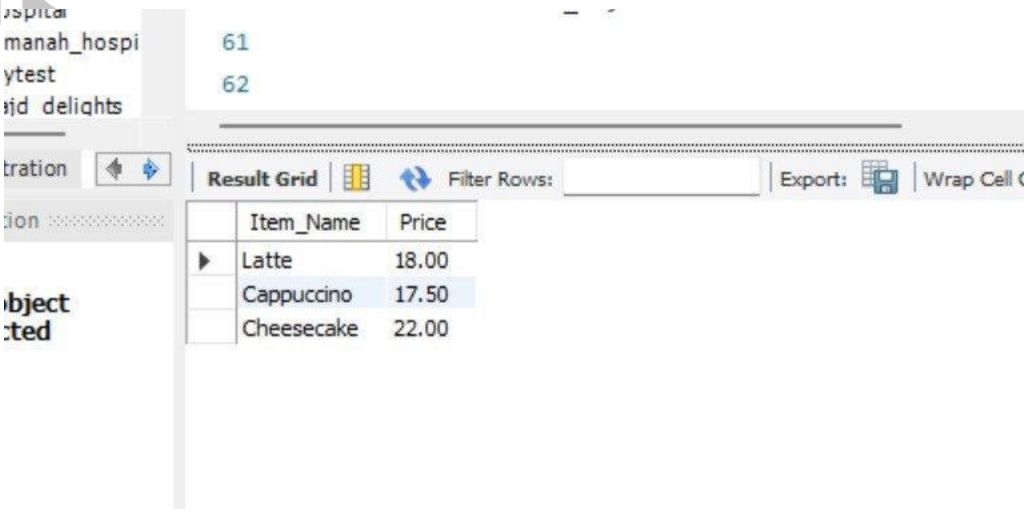
Screenshot 7: Sample Data (Table 3)



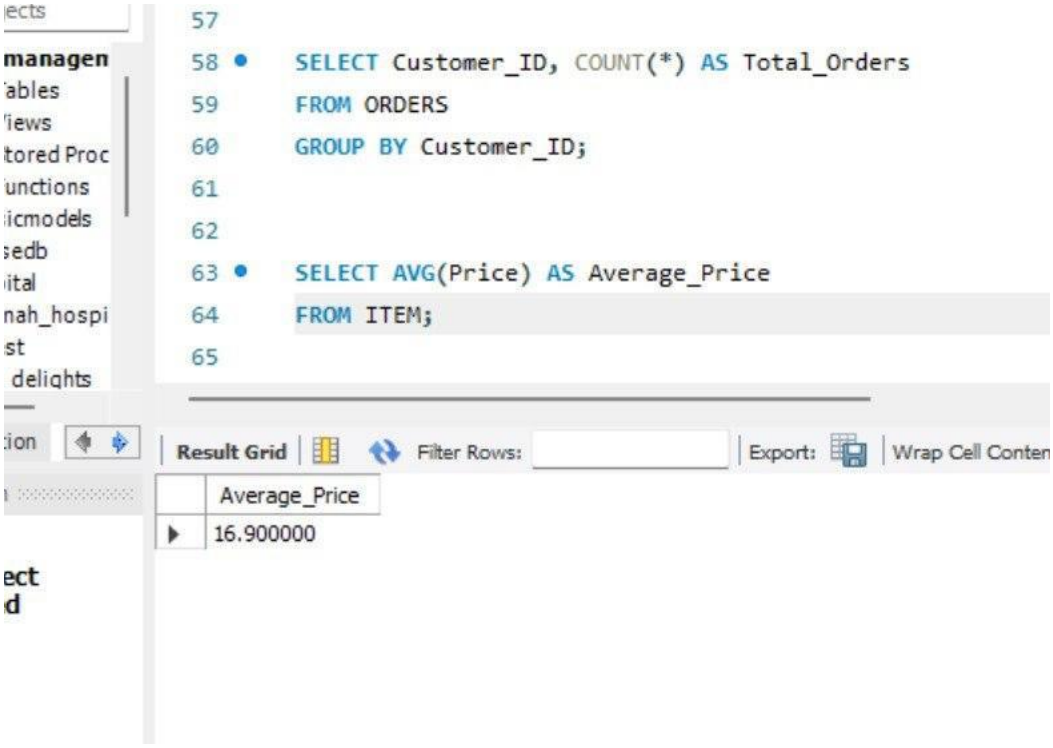
Screenshot 8: Basic SELECT Query 1 Result



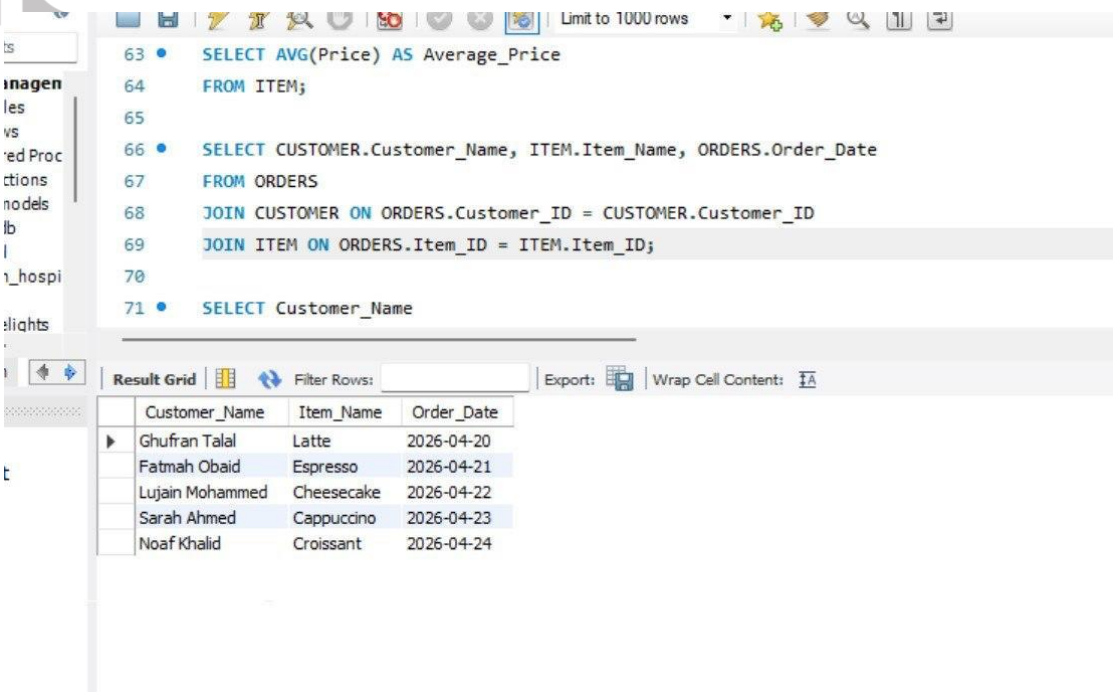
Screenshot 9: Basic SELECT Query 2 Result



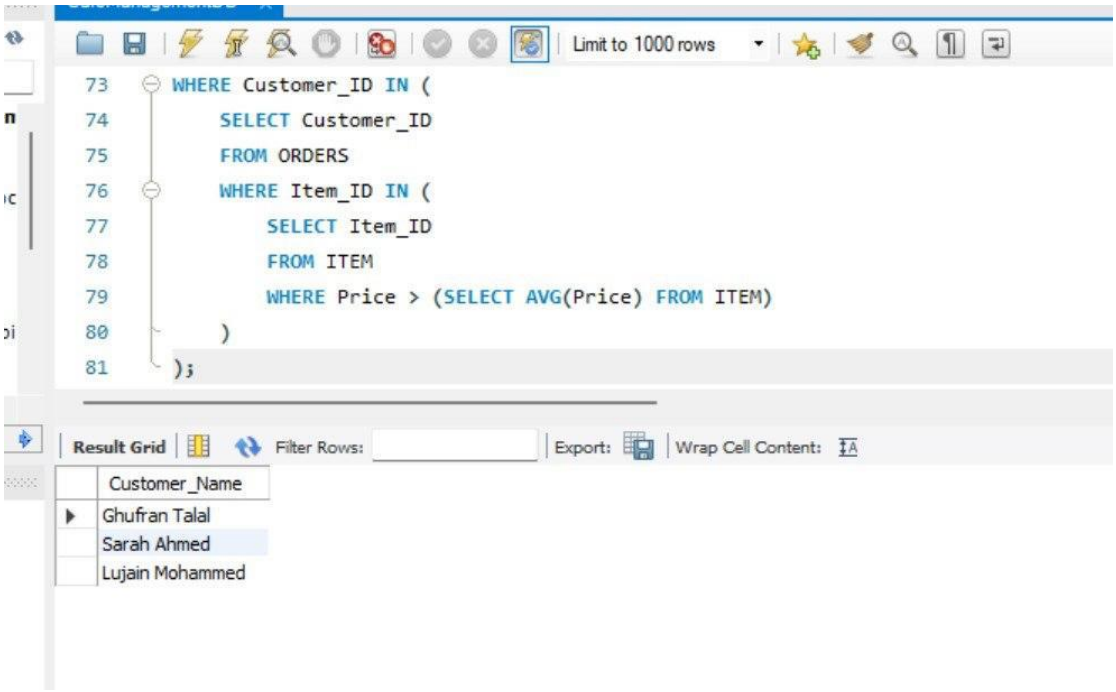
Screenshot 10: Aggregate Function Result



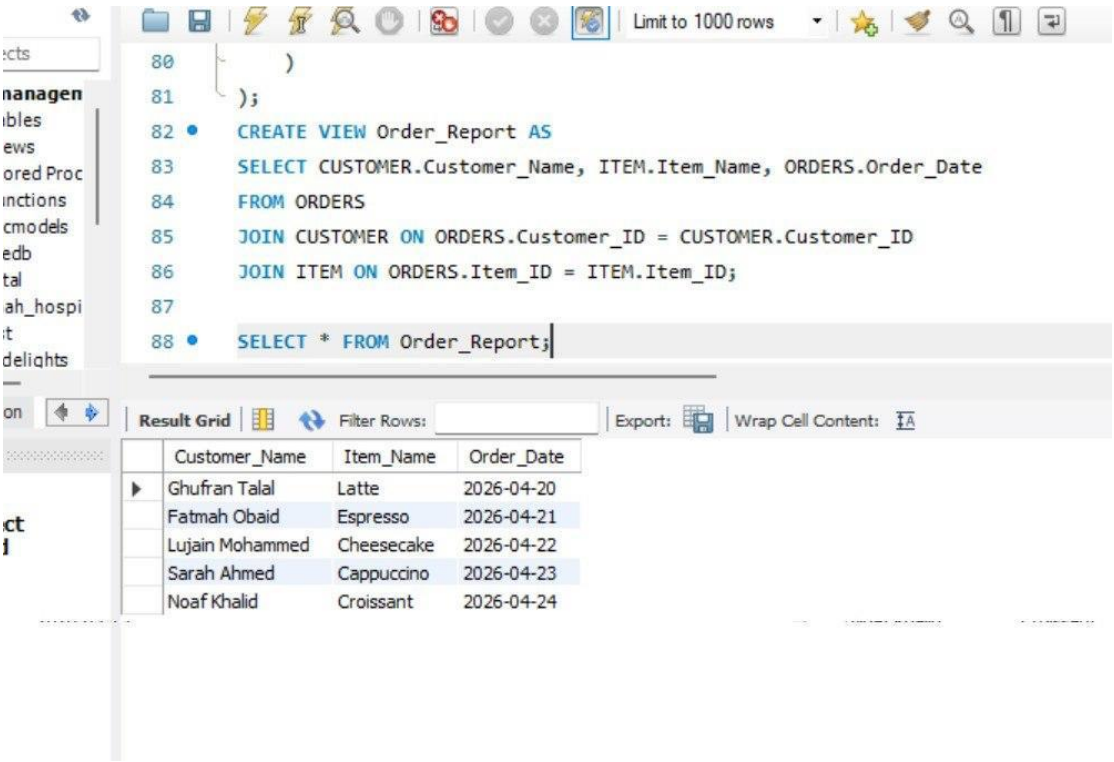
Screenshot 11: JOIN Query Result



Screenshot 12: Subquery Result



Screenshot 13: View Result



Conclusion

In conclusion, this project successfully demonstrated the complete lifecycle of a database system, starting from conceptual design using ER Diagrams to logical implementation in MySQL Workbench. By developing the Cafe Management System, we were able to apply core database concepts such as normalization, data integrity constraints, and complex SQL queries.

The system efficiently manages customers, menu items, and orders, providing a scalable foundation for cafe operations. Through this practical experience, we have gained a deeper understanding of how relational databases function in real-world scenarios and how they can be used to generate meaningful business reports through Views and Joins.



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